7.1 An Introduction to A Level Organic Chemistry (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.1 An Introduction to A Level Organic Chemistry (A Level Only)
Difficulty	Easy

Time allowed: 40

Score: /33

Percentage: /100

Question 1a

The structural formulae of four different organic compounds are shown in Table 1.1.

Table 1.1

Structural formula	IUPAC name
$C_6H_5CH_3$	
$CH_3CH_2CH_2CH_2CHO$	
CH_3NHCH_3	
$CH_3CH(NH_2)CO_2H$	

Complete Table 1.1 by writing the systematic name for each compound.

[4 marks]

Question 1b

Draw the fully displayed formula for butanoyl chloride.

butanoyl chloride

[1 mark]

Question 1c

Butanoyl chloride can react with water to form butanoic acid. State the type of mechanism that this reaction involves.

[1 mark]

Question 2a

This question is about benzene.

Draw the displayed formula for benzene.

[2 marks]

Question 2b

State the bond angle in the planar regular hexagon structure of benzene.

[1 mark]

Question 2c

State the type of hybridisation of the carbon atoms in benzene.

[1 mark]

Question 3a

This question is about isomerism in different acids.

Hydroxypropanoic acid has two position isomers.

i)
Draw and name the two position isomers of hydroxypropanoic acid.

[1]

ii)
Identify which isomer is optically active.

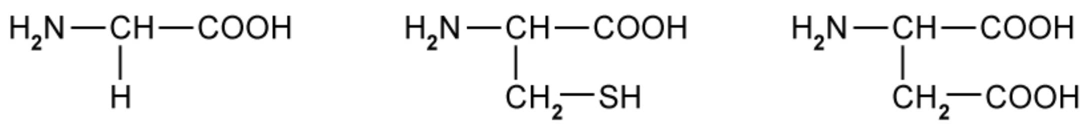
[1]

[3 marks]

Question 3b

There are 20 amino acids used, by the cells in the human body, for protein synthesis. There is only one amino acid that is not optically active.

Three amino acids are shown in Fig. 3.1.



Glycine | Cysteine | Aspartic acid

Fig. 3.1

Identify the amino acid that does **not** have any effect on plane polarised light.

[1 mark]

Question 3c

Serine, shown in Fig. 3.2, exhibits optical activity.

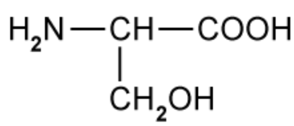


Fig. 3.2

Draw three-dimensional representations of the two isomers to show the relationship between them.

[2 marks]

Question 4a

Benzene can react with aluminium bromide, $AlBr_3$, and bromoethane, CH_3CH_2Br, to form ethylbenzene.

Write the equation for the formation of the $CH_3CH_2^+$ species.

[1 mark]

Question 4b

Name the mechanism for the formation of ethylbenzene from benzene and aluminium bromide.

[1 mark]

Question 4c

Using the $CH_3CH_2^+$ electrophile, draw the mechanism for the conversion of benzene into ethylbenzene. Include all necessary curly arrows and charges.

[3 marks]

Question 4d

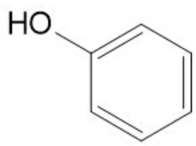
State the type of reaction that occurs in part **(c)**.

[1 mark]

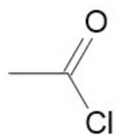
Question 5a

Compounds **K**, **L** and **M** are organic compounds with different properties.

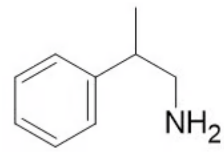
K



L



M



Give the systematic name of compounds **K**, **L** and **M**.

compound **K** ....

compound **L** ....

compound **M** ....

[3 marks]

Question 5b

Complete Table 5.1 to identify if compounds **K**, **L** and **M** from part **(a)** are aliphatic or aromatic.

Table 5.1

	Aliphatic	**Aromatic**
compound **K**		
compound **L**		
compound **M**		

[2 marks]

Question 5c

i)

State what is meant by **structural isomerism** and give **three** specific types of structural isomerism.

[2]

ii)

State what is meant by **stereoisomerism** and give **two** specific types of structural isomerism.

[2]

[4 marks]

Question 5d

Draw three-dimensional diagrams to show the stereoisomers of compound **M**.

isomer 1

isomer 2

[2 marks]

Model Answers: Easy

1a

a) The completed table is:

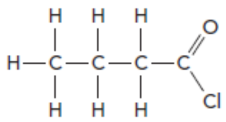
Structural formula	IUPAC name
$C_6H_5CH_3$	methylbenzene; [1 mark]
$CH_3CH_2CH_2CH_2CHO$	pentanal; [1 mark]
CH_3NHCH_3	dimethylamine; [1 mark]
$CH_3CH(NH_2)CO_2H$	2–aminopropanoic acid; [1 mark]

[Total: 4 marks]

- $C_6H_5CH_3$
  - C_6H_5 tells us that this is a benzene ring
  - The 5 hydrogen atoms in the structure tells us that 1 hydrogen atom has been substituted, in this case for a CH_3 group
  - Therefore it is methylbenzene
- $CH_3CH_2CH_2CH_2CHO$
  - The CHO group at the end of the molecule indicates the compound is an aldehyde
  - There are 5 carbon atoms in the straight chain, therefore the name is pentanal
- CH_3NHCH_3
  - The NH atom indicates that this is an amine
  - There are two CH_3 groups bonded to the nitrogen atom, dimethyl- should be in the name
  - Therefore the name is dimethylamine
- $CH_3CH(NH_2)CO_2H$
  - This compound is an amino acid
  - There are 3 carbon atoms in the chain with a carboxylic acid group, CO_2H
    - This group takes priority so is named last -propanoic acid
  - There is also a NH_2 group, amine group bonded to the second carbon
    - Therefore the name is 2–aminopropanoic acid

1b

b) The fully displayed formula for butanoyl chloride is:



- Correct displayed formula of butanoyl chloride; [1 mark]

[Total: 1 mark]

- Butanoyl chloride is an acyl chloride molecule so will contain COCl group at the end of the chain
- The carbon in the COCl group is part of the carbon chain so take note of this when drawing or naming acyl chlorides

1c

c) The type of mechanism is:

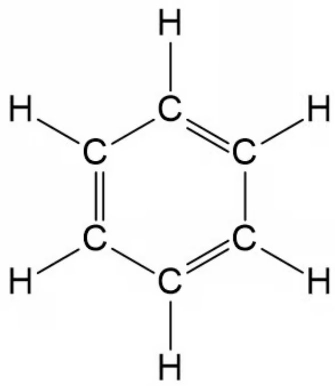
- Addition-elimination; [1 mark]

[Total: 1 mark]

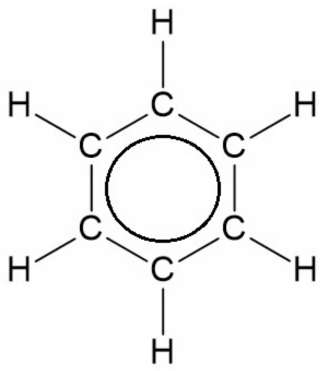
- You are asked for the type of **mechanism** that occurs in this reaction
- A water molecule adds across the C=O bond
- A hydrochloric acid (HCl) molecule is eliminated
    - Therefore the mechanism is addition-elimination
- The reaction overall is hydrolysis

2a

a) The displayed formula of benzene is:



OR



- Correct displayed formula for benzene; [1 mark]

[Total: 1 mark]

- The delocalisation of electrons in benzene is represented by drawing a circle within the hexagon
  - This increases stability
- **Remember:** The fully displayed formula should show all of the atoms and all of the bonds

2b

b) The bond angle in benzene is:

- 120°; [1 mark]

[Total: 1 mark]

- Benzene is a planar regular hexagon with bond angles of 120°
- A molecule with three bonded regions will be planar and have three bond angle of 120°, the same occurs in carbonyls
  - Square planar molecules have four bonded regions and have a bond angle of 90 °
- All the bonds are identical due to the delocalisation of electrons

2c

c) The type of hybridisation of the carbon atoms is:

- sp^2; [1 mark]

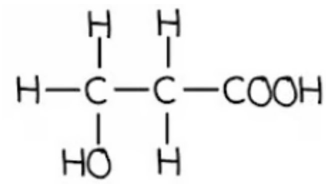
[Total: 1 mark]

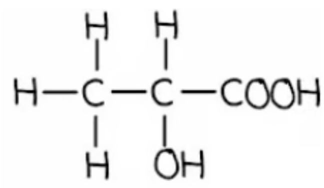
- Benzene and other aromatic compounds contain sp^2 hybridised carbons
    - Two of their p orbitals have mixed with an s orbital
- This means that each carbon atom in benzene and other aromatic compounds has one p orbital

3a

a)

i) The two position isomers of hydroxypropanoic acid are:

- 

  AND
  3-hydroxypropanoic acid; [1 mark]
- 

  AND
  2-hydroxypropanoic acid; [1 mark]

ii) The isomer that is optically active is:

- 2-hydroxypropanoic acid; [1 mark]

[Total: 3 marks]

- Part (i) is a test of your ability to draw compounds from names
- The alcohol group cannot go on the carboxylic acid carbon, therefore, it must go on carbon-2 or carbon-3
- Part (ii) requires you to check your structures for a carbon with four different atoms or groups of atoms

3b

b) The amino acid that does **not** have any effect on plane polarised light is:

- Glycine / Gly; [1 mark]

[Total: 1 mark]

- **Remember:** Optically active means that there is a carbon with four different atoms or groups of atoms attached
- Optically active compounds will have an effect on plane polarised light
  - One of the optical isomers will rotate the plane of polarised in the **clockwise** direction
  - Whereas the other isomer will rotate it in the **anti-clockwise** direction
- This question is asking for the amino acid that is not optically active
  - Therefore, you are looking for a carbon that has at least two of the same atoms or groups of atoms

b) The amino acid that does not have any effect on plane polarised light is:

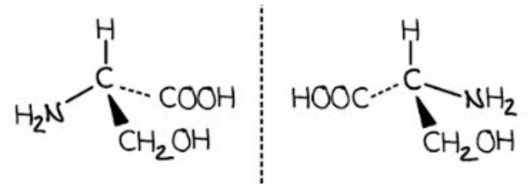
- Glycine / Gly; [1 mark]

[Total: 1 mark]

- **Remember:** Optically active means that there is a carbon with four different atoms or groups of atoms attached
- Optically active compounds will have an effect on plane polarised light
  - One of the optical isomers will rotate the plane of polarised in the **clockwise** direction
  - Whereas the other isomer will rotate it in the **anti-clockwise** direction
- This question is asking for the amino acid that is not optically active
  - Therefore, you are looking for a carbon that has at least two of the same atoms or groups of atoms
    - Glycine has two hydrogens attached to the central carbon

3c

c) The three-dimensional representations of the two isomers of serine are:



- Correct 3-dimensional structure of the left-hand isomer; [1 mark]
- Correct optical isomer of the right-hand isomer; [1 mark]
  Allow other variations as long as the correct groups are attached to the chiral carbon and they are mirror images

[Total: 2 marks]

- The chiral carbon in serine has 4 different groups attached to it
  - Hydrogen atom, amine group, carboxylic acid group and a CH_2OH group
- When drawing the 3-dimensional shape, the chiral carbon should be the central atom
- There should be four bonds coming off the chiral carbon
  - Two standard bonds (straight lines)
  - One wedge bond coming forwards from the plane of paper
  - One dotted bond going back from the plane of paper

- The optical isomer is the mirror image of the molecule drawn

4a

a) The equation for the formation of the $CH_3CH_2^+$ species is:

- $CH_3CH_2Br + AlBr_3 \rightarrow CH_3CH_2^+ + AlBr_4^-$; [1 mark]

[Total: 1 mark]

- Aluminium bromide, $AlBr_3$ and aluminium chloride, $AlCl_3$ can both act as catalysts in the formation of an electrophile in Friedel-Crafts alkylation and acylation
  - **Careful:** Make sure that the halogen atoms match, e.g. $AlBr_3$ for reactions with bromoalkanes and $AlCl_3$ for reactions with chloroalkanes

4b

b) The mechanism for the formation of ethylbenzene from benzene and aluminium bromide is:

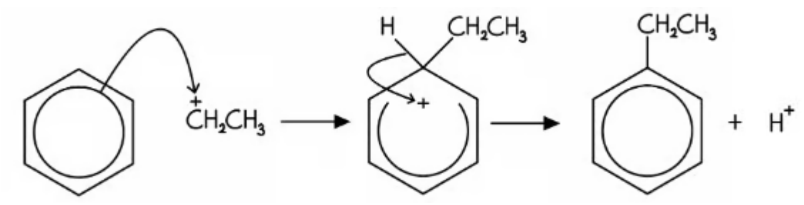
- Electrophilic substitution; [1 mark]

[Total: 1 mark]

- If you see a mechanism which involves a hydrogen on the benzene ring being substituted, you can be sure that the mechanism is an electrophilic substitution
- The overall general mechanism is the same for all electrophilic substitution reactions, which makes it more manageable to learn

4c

c) The mechanism for the formation of ethylbenzene is:

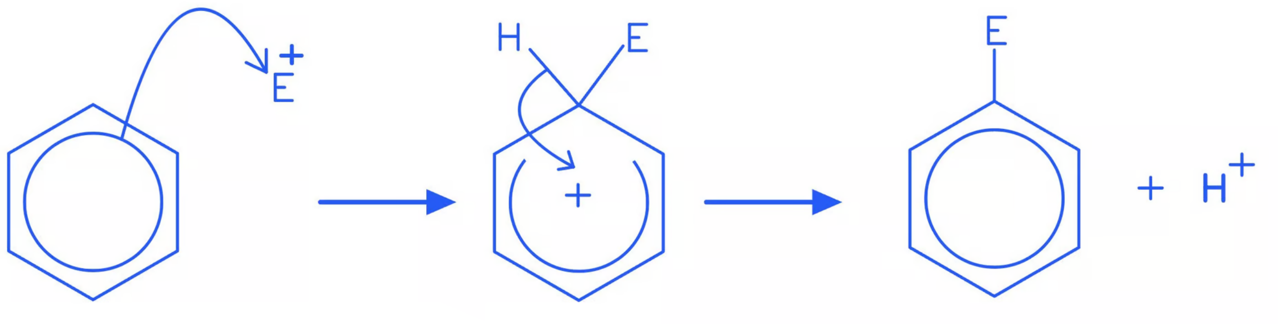


- Curly arrow from the ring inside benzene to the C+ of the CH3CH2+ ion; [1 mark]
- Correct intermediate structure; [1 mark]

- Curly arrow from the C-H bond back to the + within the ring; [1 mark]

[Total: 3 marks]

- There are extra marking points for electrophilic substitution mechanisms:
  - The + must be on the correct carbon in the $CH_3CH_2^+$ electrophile
  - The first arrow must start touching the ring, not start inside the ring
  - The horseshoe in the ring must be centred about carbon 1 and not extend past carbon 2 or 6
- All electrophilic substitution mechanisms follow the same general pattern:



- The electrophile bonds to the benzene ring, forming an intermediate with a partially delocalised electron system - shown as a horseshoe with + inside
- The intermediate loses a proton, restoring the delocalised electron system

4d

d) This type of reaction is:

- (Friedel-Crafts) alkylation; [1 mark]

[Total: 1 mark]

- It can be easy to confuse the type of reaction and the mechanism
  - Friedel-Crafts alkylation involves the electrophilic substitution of an alkyl group with a hydrogen atom to bond onto the benzene ring
  - Friedel-Crafts acylation involves the electrophilic substitution of an R-C=O group with a hydrogen atom to bond onto the benzene ring

5a

a) The systematic name of compounds **K**, **L** and **M** are:

- compound **K** = phenol; [1 mark]
- compound **L** = ethanoyl chloride; [1 mark]
- compound **M** = 2–phenylpropylamine; [1 mark]

[Total: 3 marks]

- You should know that compound **K** is phenol
- Compound **L** is an acyl chloride
    - These have the suffix -oyl chloride
    - The carbon chain:
        - Is 2 carbons long, which means that the name is eth- based
        - Contains only single bonds, which means that the name contains -an-
    - Overall, you have eth + an + oyl chloride = ethanoyl chloride
- Compound **M** is more complicated
    - There is a phenyl ring attached to carbon-2 of a carbon chain, which means that the name contains 2–phenyl
    - The carbon chain is 3 carbons long **AND** has an amine group at the end, which means that the name contains propylamine
    - Overall, you have 2–phenyl + propylamine = 2–phenylpropylamine

5b

b) The completed Table 5.1 identifying whether compounds **K**, **L** and **M** are aliphatic or aromatic is:

	Aliphatic	**Aromatic**
compound **K**	x	✓
compound **L**	✓	x
compound **M**	✓	✓

- Compounds **L** and **M** identified as aliphatic; [1 mark]
- Compounds **K** and **M** identified as aromatic; [1 mark]

[Total: 2 marks]

- **Remember:** Aromatic compounds contain a benzene ring
    - Aliphatic compounds do not contain benzene rings
    - Aliphatic compounds are most commonly carbon chains although they can

sometimes be cyclic as well, e.g. cyclohexane

5c

c)

i) **Structural isomerism** is:

- Where compounds have the same molecular formula but different structural formulae; [1 mark]

Three specific types of structural isomerism are:

- (Branched) Chain
  AND
  Positional
  AND
  Functional group; [1 mark]

Three specific types of structural isomerism are:

- (Branched) Chain
  AND
  Positional
  AND
  Functional group; [1 mark]

ii) **Stereoisomerism** is:

- Where compounds have the same molecular formula and structural formulae but a different arrangement of atoms in space; [1 mark]

Two specific types of structural isomerism are:

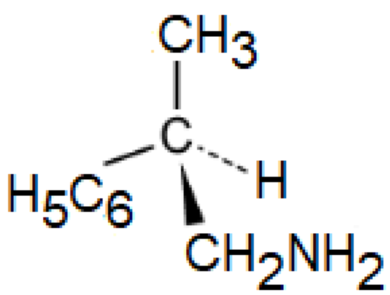
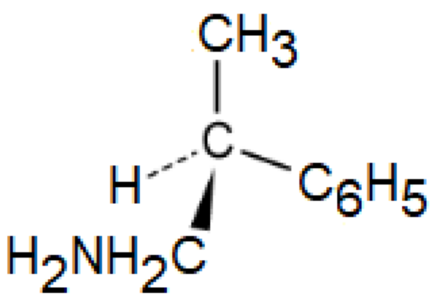
- Geometric(al)
  AND
  Optical; [1 mark]

[Total: 4 marks]

- You need to know the definitions of structural isomerism and stereoisomerism as well as be able to work with each type of isomer that they describe

5d

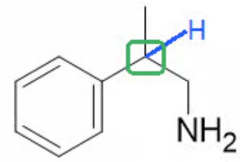
d) The three-dimensional diagrams showing the stereoisomers of compound **M** are:

isomer 1	isomer 2
	

- Correct structure of one optical isomer; [1 mark]
- Mirror image of the first optical isomer; [1 mark]

[Total: 2 marks]

- Stereoisomers that require three-dimensional diagrams drawing using the template provided are optical isomers
- **Remember:** Optical isomers have a chiral centre, i.e. one carbon atom that has four different atoms or groups of atoms attached
- Take your time identifying the chiral centre and don't forget that hydrogen atoms may not be displayed in the structure that you are working with



- This means that there is a CH_3 group, hydrogen atom, CH_2NH_2 atom and a benzene ring to place on the isomer templates
  - **Careful:** The formula of the benzene ring is **not** C_6H_6 because one of the hydrogen atoms has been substituted